Saturated Superfluid Vacuum IV
Gravity as Update-Capacity Gradient

Stig Norland

Draft

Abstract

This paper develops gravity in the Saturated Superfluid Vacuum (SSV) framework as the bending of motion through gradients of local time delay. It builds on Paper II, where gravity enters the force sector as the interference of the pressure waves that every massive breather radiates into the vacuum, and on Paper III, where physical time is identified with the local rate of change of the order parameter Ψ — the medium's local *update capacity*. The chain is short: the interference of radiated waves consumes update capacity and so slows local time; that slowing is uneven and therefore has a gradient; and any motion through the gradient is bent toward the region of greatest delay. That bending is gravity. From a single interference potential Φ follow free fall, gravitational time dilation, light deflection, and redshift, each at leading order in Φ/c^2. The treatment is weak-field throughout: it delivers the time sector of gravity in full — including the time-delay (Einstein 1911) half of light deflection — with the spatial sector and the recovery of full general relativity left to Paper VII-b.

Contents

1 Introduction

Gravity is usually approached either as a field to be quantised or as the curvature of a spacetime taken as fundamental. This paper takes neither route. In the Saturated Superfluid Vacuum (SSV) the vacuum is a real medium — a saturated compressible superfluid described by an order parameter Ψ — and the question is not how to quantise a pre-existing gravitational field, but how gravitational behaviour arises at all if the medium's local rate of change is finite and depends on how heavily the medium is loaded.

Two earlier papers supply the foundation. Paper II places gravity in the force sector: every massive defect is a breather, a localised oscillation of Ψ, and it radiates a longitudinal pressure wave into the medium; the time-averaged interference of two such waves is the gravitational interaction, and matching it to the observed inverse-square law fixes Newton's constant G from the medium parameters. Paper III identifies physical time with the local rate of change of Ψ: time is not a backdrop but a measure of how fast the medium is updating. The present paper takes the mechanism introduced in Paper II and develops it in its own right, on the footing Paper III provides.

The thesis is a single chain. The interference of radiated waves slows the local rate of change of Ψ — it imposes a time delay. The delay is uneven, strongest near a mass and fading with distance, so across space it is a gradient. Any motion through that gradient is bent toward the side where time runs slower. That bending is gravity. A single potential Φ summarises the delay, and from it follow free fall, gravitational time dilation, light deflection, and redshift.

The treatment here is weak-field, exact at leading order in Φ/c^2, and deliberately confined to one sector. It delivers the *time* sector of gravity in full: gravitational time dilation, redshift, free fall, and the time-delay part of light deflection — the last equal to Einstein's 1911 value $2GM/(bc^2)$. Light deflection draws equally on a second, spatial sector; that sector, the effective metric to all orders, the post-Newtonian coefficients, and the dissolution of the quantum-gravity problem are the subject of Paper VII-b, to which §8 hands off. This is a division of labour, not a gap — the present paper is complete for what it claims. Galactic-scale applications of the same mechanism are treated in Papers VI-a and VI-b. Section 2 formalises update capacity; §3 sets out the physical picture; §§4–5 make it quantitative; §6 develops the refraction geometry of path bending; and §§7–9 treat universality, the absence of a graviton, the bridge to emergent geometry, and the connection to the recording medium of Paper III.

2 Update Capacity and Local Change-Rate

Paper III identifies physical time with the local rate of change of Ψ, and entropy with the persistent wake-complexity that change leaves behind. This section makes the rate itself the object of study and gives it a name: the medium's local *update capacity*.

2.1 Updates, and the capacity to make them

The medium does not evolve smoothly and for free. It advances through discrete interactions, and the rest of this section depends on being precise about what one such interaction is. An update is not any change of Ψ whatever; it is a *wake-writing event* in the sense of Paper III — a change that deposits irrecoverable wake. Paper III already draws this line: stationary zero-point fluctuations are excluded from its event count, because by construction they leave no net wake. An update is a change that does leave wake.

Two facts about updates then fix everything that follows. First, each update takes a finite, non-zero time: the medium cannot rewrite its state in no time at all. Second, because each update costs time, only a bounded number of them fit into any given duration. That bound is the update capacity: the number of wake-writing changes the medium can make, per unit time, at a given place. It is the rate-side of Paper III — where Paper III counts the wake updates leave behind, read backwards as entropy, here we count the updates themselves, forwards, as they happen.

Because physical time *is* the local rate of change of Ψ (Paper III, $d\tau \propto \mathcal{R}[\Psi]\, dt$), the count of updates a process performs is its proper time — a clock that has completed N internal updates has aged by N ticks, and there is nothing else for its age to be. Update capacity is therefore not a loose analogy for proper time; it is proper time, counted as a rate.

2.2 The ceiling

There is a fastest possible update rate, and it can be argued rather than assumed. The argument has two steps.

First, the smallest thing that can carry an update is the grain $a_p = \hbar/(m_p c) \approx 2.1 \times 10^{-16}$ m, the proton Compton radius. An update must be a wake-writing event; a change localised to less than a_p writes no persistent wake, because below a_p nothing is stable enough to persist (§7). A sub-grain change therefore stands on exactly the footing of the zero-point fluctuations Paper III already excludes from the event count. The finest thing that counts as an update is grain-sized.

Second, a grain cannot complete a change faster than a signal can cross it. Nothing propagates faster than the medium's speed c, and a grain cannot register a new state before a signal has had time to traverse it. The minimum update time is therefore the grain's light-crossing time,

$$t_0 = \frac{a_p}{c} \approx 7.0 \times 10^{-25}\,\text{s}, \tag{1}$$

and the update-capacity ceiling — the most updates per unit time the medium can make anywhere — is its reciprocal,

$$N_0 = \frac{c}{a_p} = \frac{m_p c^2}{\hbar} \approx 1.4 \times 10^{24}\ \text{s}^{-1}. \tag{2}$$

This is the causal content of the photon: a photon crosses one grain in exactly t_0, updating the grains along its track at the rate N_0 and no faster. A photon is the medium being driven at its ceiling.

The ceiling is fixed twice over, and the two ways agree. Geometrically it is c/a_p, the light-crossing rate of the grain. Energetically it is $\omega_p = m_p c^2/\hbar$, the Compton angular frequency of the heaviest stable defect, the proton: the medium updates precisely as fast as its fastest stable resident oscillates. A process demanding updates faster than N_0 would be asking the medium to resolve structure finer than its own grain. The geometric and energetic routes to N_0 are one fact stated twice.

Open: the order-unity factor

The argument fixes $N_0 \sim c/a_p$ and identifies the grain with the proton Compton radius. What it does not fix is the dimensionless factor of order unity: whether the minimum update length is exactly a_p, or a_p times a number that a wake-deposition calculation in the LogSE would settle. The identity $c/a_p = \omega_p$ is exact only if that factor is exactly one; the present argument establishes that the geometric and energetic ceilings agree up to that single ambiguity.

2.3 Two uses of one budget: kinematic time dilation

A defect does not run at the ceiling N_0 unless it is the proton at rest. Its own demand is $\omega_0 = mc^2/\hbar \leq N_0$ — the rate at which it must update simply to remain itself, to maintain its internal breather oscillation. At rest in undisturbed vacuum it meets that demand in full, with every update going to internal oscillation.

A defect in motion cannot. Translation is itself updating: to carry its pattern through the medium a defect must update the grains along its path, and those updates come from the same account that runs its internal oscillation. Internal interaction and translation are two uses of one budget.

Take the budget magnitude fixed at the rest value ω_0 and the two uses as orthogonal draws on it — internal and translational updates combining in quadrature:

$$N_{\text{int}}^2 + N_{\text{trans}}^2 = \omega_0^2, \qquad \frac{N_{\text{trans}}}{\omega_0} = \frac{v}{c}. \tag{3}$$

The internal oscillation — the defect's clock — then runs at

$$N_{\text{int}} = \omega_0\sqrt{1 - v^2/c^2}, \tag{4}$$

slow by exactly the special-relativistic factor, and slow for a mechanical reason: updates spent on getting somewhere are updates not spent on ticking.

Honesty requires naming what (3) is. The fixed-magnitude, quadrature rule is not derived here from the medium's dynamics; it is the kinematic face of the *emergent* Lorentz invariance of the medium's excitations. This is not a weakness peculiar to update counting. The logarithmic Schrödinger equation is first order in time — it is Galilean-invariant — so special relativity is not native to it and cannot be: it must emerge, exactly as Lorentz invariance emerges for the quasiparticles of the analogue-gravity systems on which this framework draws [2, 3], with the medium's signal speed c as the invariant. Equation (3) states that the update budget is the invariant total this emergent structure holds fixed — the same structure that §8 expresses geometrically as the acoustic metric. The quadrature and the acoustic metric are one emergent object; deriving it from the LogSE is a single open problem, shared with the emergent-geometry programme of Paper VII-b. What §2.3 establishes is the identification: kinematic time dilation and the update budget are the same thing. The next subsection collects the dividend, because the same budget can be diminished a second way.

2.4 Loading, availability, and gravitational time dilation

The kinematic split assumed undisturbed vacuum: the budget ω_0 was available in full, and only its division was at issue. Where the medium is not undisturbed, the budget itself is reduced.

When the medium at a location already carries the interfering pressure waves that surrounding matter radiates into it (§4), an embedded process does not find the medium idle. The mechanism is the interference cross-term of §4, and it is computed, not assumed, in §5: a clock placed in that wave acquires the cross-term with it, and its energy is shifted by the interaction energy $m\Phi$. A process's update rate is its energy divided by $\hbar$ — that is the whole content of $\omega_0 = mc^2/\hbar$ — so a shift in energy is, with nothing further assumed, a shift in update rate. The clock realises not ω_0 but

$$N_{\text{realised}} = \frac{mc^2 + m\Phi}{\hbar} = \omega_0 \left(1 + \frac{\Phi}{c^2}\right). \tag{5}$$

Define the *availability* A as the fraction of its demand a clock actually realises:

$$A \equiv \frac{N_{\text{realised}}}{\omega_0} = \frac{d\tau}{dt} = 1 + \frac{\Phi}{c^2}. \tag{6}$$

This is not a new quantity and not a metaphor. A is $d\tau/dt$, and its value is the cross-term calculation of §5; "the interference consumes updates" means precisely that the cross-term lowers the embedded oscillator's energy, and energy is update rate. The defect mass m has cancelled — the interaction energy is $m\Phi$, the rest energy mc^2, their ratio Φ/c^2 whatever the clock is made of — so the universality of gravitational time dilation is not a separate postulate but a consequence of every clock spending one shared kind of budget.

The two reductions can now be read together. A clock loses update rate when it moves — the kinematic split, (4) — and it loses update rate when the medium around it is loaded — the gravitational reduction, (6). Both are the one update budget being diminished; special-relativistic and gravitational time dilation are not two phenomena but two ways of drawing down the same account. What remains is to compute Φ from the loading and to show that it varies across space: a quantity that varies across space has a gradient, and the gradient of the availability is the time-delay gradient of §3 — motion across it is gravity.

3 The Physical Picture

Gravity is the gradient of a time delay — and that delay is built by the interfering pressure waves that every mass radiates into the vacuum. Three steps connect those two statements: the interference of waves slows local time; that slowing is uneven, so across space it is a gradient; and any motion through that gradient is bent toward the slower side. The bending is gravity. This section sets out the chain; §§4–6 make each step quantitative.

In Paper I every massive defect is a breather: a localised region of Ψ oscillating at its Compton frequency. A breather does not sit inertly in the medium. It radiates a longitudinal pressure wave — a travelling density modulation $\delta\rho(\mathbf{x}, t)$ — outward at the medium's sound speed c, with amplitude falling as $1/r$ from spherical dilution. The region surrounding a mass is therefore not a patch of medium with some altered static property; it is a region permeated by that radiated wave.

Interference slows local time. In SSV, time is not a backdrop through which things move; it is the local rate at which Ψ changes (Paper III; §2). Any process — a clock, a particle's internal oscillation, a light wave — advances by changing Ψ, and in undisturbed vacuum it advances at the medium's full rate. A process embedded in a breather's radiated wave is not alone: the medium around it is already oscillating with that wave, and the process must run *through* it. The two interfere. Through the medium's logarithmic nonlinearity this interference has a definite, one-signed effect (§4): it lowers the embedded process's rate of advance, so that fewer of its cycles complete per unit coordinate time. Because physical time *is* that rate of advance, the process runs slow — and the more intense the radiated wave, the greater the slowing. This is the time delay.

The delay is uneven, so it has a gradient. The radiated wave dilutes as $1/r$, so the interference is most intense at the mass and fades with distance. Local time therefore runs slowest at the mass and recovers the full vacuum rate far away. The delay is not a uniform offset added everywhere alike; it varies smoothly from place to place, and a quantity that varies across space has a gradient — here one that points toward the mass and steepens as the mass is approached.

The gradient bends motion, and that is gravity. A gradient of time delay deflects everything that crosses it. Anything that moves is a wave or an oscillation advancing through the medium — a light ray carrying its front forward, a body carrying its Compton oscillation. Where local time is uneven, the side of that advance nearer the mass keeps slower pace than the side farther away; the front pivots, wheeling toward the slow side, exactly as a wavefront refracts toward a slower medium or a rank of marchers wheels toward the flank taking shorter steps. The trajectory curves toward the region of greatest time delay — toward the mass. A light ray grazing the Sun bends this way: not by passing through "denser matter" but by advancing through the Sun's radiated wave, which runs its near side slow. Even a body at rest is internally a pulsating breather, and its oscillation runs slow on the side facing the mass; that tilt of its internal phase is itself momentum directed toward the slow side, and the body falls. An object never feels the distant mass — it responds only to the local slope of the time delay where it sits. That slope, not the mass, is the source of gravity; the mass merely shapes it.

The time-averaged interference is summarised by a single potential $\Phi(\mathbf{x})$, and from it follow the four observed gravitational phenomena, derived in turn below:

- **free fall** — a test mass drifts down the potential gradient (§6);
- **time dilation** — a clock deeper in the potential ticks slow (§5);
- **light deflection** — a light wave's front pivots in the potential gradient (§5, 6);
- **gravitational redshift** — a wave climbing out of the potential loses frequency (§5).

The acceleration of free fall is $-\nabla\Phi$ and the fractional slowing of clocks is Φ/c^2. The interference cross-term of §4 builds Φ from the radiated waves; the sections after it read off the consequences.

4 The Interference Cross-Term

The mechanism is already contained in the LogSE. The medium's nonlinear self-interaction is the logarithmic term $b\ln(\rho/\rho_0)$ of the Paper I Lagrangian. Consider a point of the medium reached by the radiated waves of several breathers. The local density is the saturation value modulated by each wave,

$$\frac{\rho(\mathbf{x},t)}{\rho_0} = 1 + \sum_i x_i(\mathbf{x},t), \qquad x_i \equiv \frac{\delta\rho_i}{\rho_0}, \tag{7}$$

and the logarithmic energy the medium carries there is governed by $\ln(\rho/\rho_0)$. Expanding,

$$\ln\Big(1+\sum_i x_i\Big) = \sum_i x_i - \frac{1}{2}\Big(\sum_i x_i\Big)^2 + \cdots = \sum_i x_i - \frac{1}{2}\sum_i x_i^2 - \sum_{i<j} x_i x_j + \cdots \tag{8}$$

The single-wave terms x_i and x_i^2 are each breather's own self-energy. The interaction between two distinct breathers A and B is carried entirely by the cross term

$$u_{AB} \;=\; -\,b\,\rho_0\,x_A\,x_B. \tag{9}$$

This is an *interference* term: the product of two waves, not a property of either alone. Three features of (9) fix the character of gravity.

It survives averaging only for coherent waves. Each x_i oscillates at its breather's Compton frequency. The observable interaction is the time average $\langle u_{AB}\rangle = -b\rho_0\langle x_A x_B\rangle$, and $\langle x_A x_B\rangle$ vanishes unless the two waves are phase-correlated. Incoherent waves do not gravitate toward one another; coherent ones do. All matter breathers are eigenmodes of the one medium and lock in phase through it (a resonance condition; see Paper II), so $\langle x_A x_B\rangle > 0$ generically.

Its sign is fixed by the concavity of the logarithm. The cross term carries the minus sign of the $-\frac{1}{2}(\sum x)^2$ expansion — a direct consequence of ln being concave. For phase-correlated waves $\langle x_A x_B\rangle > 0$, so $\langle u_{AB}\rangle < 0$: the interaction energy is lowered where the waves overlap in phase. *Gravity is attractive because the medium's nonlinearity is logarithmic, and the logarithm is concave.*

It is bilinear, so the potential goes as $1/r$. Equation (9) is linear in each wave. Breather B, in its own localised volume, couples to breather A's wave evaluated at B's location, where spherical dilution has reduced it to $x_A \propto m_A/r$; its own wave has $x_B \propto m_B$. The time-averaged interaction energy is therefore

$$\langle u_{AB}\rangle \;\propto\; -\frac{m_A m_B}{r}, \tag{10}$$

the Newtonian form. The gravitational potential Φ is this interference energy per unit test mass; the force is $-\nabla\Phi \propto m_A m_B/r^2$. The inverse square is the gradient of a diluted wave, not a separately postulated law.

The expansion fixes the *structure* of Φ — interference only, attractive, $1/r$. Its coefficient is fixed by the breather's acoustic output; matching the result to the observed inverse-square law yields Newton's constant, computed from the medium in Paper II.

The same time-averaged interference of pulsating waves was first analysed classically by Bjerknes [1], who obtained it as a time-averaged force $\langle F\rangle = -\rho_0\langle \dot V_1 \dot V_2\rangle/(4\pi r^2)$ between two bodies pulsating in a compressible medium. Paper II gives the historical credit, shows that $\langle \dot V_1 \dot V_2\rangle$ is exactly the cross-correlation $\langle x_A x_B\rangle$ above written in volume-velocity variables, and carries the result through to G. The derivation here is LogSE-native and needs no external pulsator: the breathers are self-sustaining eigenmodes of Ψ.

Numerical check. A direct simulation bears out the cross-term. Two localised sources are driven into pulsation in a two-dimensional logarithmic-Schrödinger medium and the time-averaged force one exerts on the other is measured. An isolated source feels none. With a second source present the force reverses *exactly* when that source's drive phase is flipped: the ratio of the in-phase to the anti-phase force is -1 to within one per cent at every separation tested, the signature of a force bilinear in the two waves, as (9) demands. In the near zone — separations well inside a sound wavelength, where retardation is negligible — two in-phase sources attract, and the force falls monotonically with separation, exactly as the concavity argument predicts.

Beyond a wavelength the time-averaged force oscillates in sign with separation, the long-known retardation behaviour of the Bjerknes force between widely separated pulsators; the near-zone regime, in which the cross-term derivation is framed, is the one tested here, and it is cleanly attractive. Appendix A records the run parameters and the convergence tests behind these numbers.

5 Time Dilation, Deflection, and Redshift

Paper II fixes the interference potential of a source mass M,

$$\Phi(r) = -\frac{GM}{r}, \tag{11}$$

with G computed from the medium; its gradient $-\nabla\Phi$ is free fall. The same Φ — with no further parameter — gives the three remaining gravitational phenomena. Each is the interference cross-term of §4 read for a different probe.

Time dilation. A clock is an oscillator: any periodic physical process carries a rest energy mc^2 and runs at $\omega = mc^2/\hbar$. Placed in the wave of a source mass, the clock's oscillation acquires the interference cross-term with that wave; its energy is shifted by the interaction energy $m\Phi$. The fractional change in tick rate is therefore

$$\frac{d\tau}{dt} = 1 + \frac{m\Phi}{mc^2} = 1 + \frac{\Phi}{c^2}, \tag{12}$$

the standard weak-field result; with $\Phi < 0$ in a well, the clock runs slow. This is not an independent derivation — it is (11) divided by c^2. The slow clock is the time delay of §3: each oscillation completes less physical change per unit coordinate time where the interference is strong, and gravity, $-\nabla\Phi$, is the gradient of that delay.

Light deflection. A light ray is itself a wave in Ψ. Passing a source mass, it propagates through the source's wave and interferes with it; the cross-term makes its local phase velocity depend on position through Φ. A transverse gradient of Φ pivots the wavefront, deflecting the ray toward the mass — the same gradient that accelerates a test mass in free fall. Light bends because a wave, like a clock, carries the interference cross-term. The refraction geometry is developed in §6.

Gravitational redshift. Redshift is time dilation read along a ray. A wave emitted at radius r_1, where the potential is Φ_1, and received at r_2, where it is Φ_2, carries the emitter's tick rate to the receiver; the fractional frequency change is

$$\frac{\Delta\nu}{\nu} = \frac{\Phi_1 - \Phi_2}{c^2}. \tag{13}$$

A wave climbing out of a well ($\Phi_1 < \Phi_2$) loses frequency. No new mechanism is invoked — only (12) applied at the two ends of a path.

Open: extension beyond leading order

Equations (11)–(12) are exact at $O(\Phi/c^2)$, the order at which the interference cross-term (9) is the leading contribution. Extension to higher orders requires keeping the cubic and higher terms of the logarithmic expansion (8) — treating the LogSE fully nonlinearly rather than at quadratic order — and confirming that the result reproduces the parametrised post-Newtonian expansion of general relativity (the PPN coefficients β, γ). This is a clean test: SSV makes a definite prediction at each order, and any departure from $\beta = \gamma = 1$ would be a structural prediction distinct from GR. The calculation is non-trivial but finite, and is carried out in Paper VII-b.

6 Path Bending and the Refraction Picture

Section 3 stated that a gradient of time delay bends every trajectory that crosses it, and §5 read light deflection off Φ as one of the four consequences. This section makes the bending a construction: it shows concretely how a region of slowed time turns a wave, and then quantifies the turn.

6.1 The wavefront construction

Begin with a picture that is not gravity at all. Ocean swell approaching a small island slows as it reaches the shallows — a water wave travels more slowly where the water is shallower. The part of each crest that enters the shallows first falls behind the part still in deep water; the crest, no longer straight, pivots; and the wave train curves inward, bending around the island. Every point of the front advances at its own local wave speed, and where that speed varies along the front, the front turns — toward the slow side. This is refraction, and it is pure geometry: it follows from nothing more than "each part of the front advances at its own local rate."

The gravitational case is this construction with one word changed. Put a mass where the island was. Around it, by the interference of §4, lies a region where the local update rate is reduced — where time runs slow. A wave crossing that region has its edge nearer the mass advancing through fewer updates per unit coordinate time than its edge farther away; by exactly the island's geometry the front pivots toward the slow edge, and the wave curves toward the mass.

The one changed word carries the whole of the physics, and it must be said plainly. The island slows the water wave by a change in the medium's static structure: the water is shallower, and the wave reads that depth directly as a local speed. Around a mass there is no such material step. The vacuum's saturation density and sound speed are the same near the mass as far from it; a probe finds no denser or thinner medium to read, and no static density blob to refract off. What is graded is the rate of time itself — the update availability A of §2, lowered where the mass's radiated waves interfere most strongly. The probe refracts in a gradient of time, not in a gradient of medium-stuff. A region where time runs slow, and nothing material besides, bends a wave that crosses it: that is the content of this section, and the island supplies only the geometry.

6.2 Particle and wave bend alike

The construction does not depend on whether light is pictured as a wave or as a particle, and in SSV that is not a coincidence to be managed but a structural fact. SSV has no point particles.

Every particle is a defect of finite extent — a breather or a vortex ring (Paper I) — and a wave is extended in its front. Everything that moves is an extended thing, and an extended thing crossing a gradient of time genuinely has a near side and a far side across which the rate differs.

As a particle, then: a defect crossing the slow-time region has its near side — the side toward the mass — advancing through fewer updates than its far side, so it rotates, and its heading turns toward the mass. As a wave: the near edge of the front lags the far edge, and the front pivots the same way. These are not two mechanisms but one, and it is mechanically literal in both readings, because in both the object really has two sides. There is no pointlike limit to worry about; in SSV nothing is pointlike.

The extent does one more thing. A reader might fear that a larger defect, with a larger near–far separation, would bend more — which would break the universality of §2.4. It does not. The defect's heading rotates at the rate (difference in advance across it) divided by (its width); and the difference in advance is itself proportional to the width, being the gradient of A times the separation. The width cancels. What is left is a path curvature fixed by the gradient of A alone — the same for a defect of any size, and the same for a wave. The equivalence principle, recovered in §2.4 as the cancellation of mass, reappears here as the cancellation of size. Its quantitative form is §6.3.

6.3 The effective index and the deflection of light

Make the wavefront construction quantitative, taking light first.

A wave's local rate of advance is the availability A of §2: where time runs slow, a wave completes fewer cycles, and advances less, per unit coordinate time. A front covers a proper distance dl in proper time dl/c; since $d\tau = A\,dt$, that is a coordinate time $dl/(cA)$, so the wave's coordinate speed is $u = cA$, and the slowed time acts as an optical medium of refractive index

$$n = \frac{c}{u} = \frac{1}{A} \approx 1 - \frac{\Phi}{c^2}. \tag{14}$$

With $A < 1$ near a mass, $n > 1$ there: light is slowed exactly as in a denser optical glass — except that what is "denser" is the rate of time, not the medium (§6.1).

The front-tilt of §6.1 now has a rate. Take a short segment of wavefront advancing along its normal; two points of it a transverse step apart advance at slightly different speeds $u = cA$, the faster end gains on the slower, and the segment tilts. The ray, normal to the front, turns with it. Dividing the tilt by the path length advanced, the ray curvature is

$$\frac{d\theta}{ds} = \frac{1}{A}\,|\nabla_\perp A| \;\approx\; \frac{1}{c^2}\,|\nabla_\perp \Phi|, \tag{15}$$

directed toward smaller A — toward greater delay, toward the mass.

For a ray grazing a point mass M at impact parameter b, with $\Phi = -GM/r$ and $r^2 = b^2 + s^2$ along the nearly straight ray, the total deflection is the curvature integrated along the path:

$$\alpha = \frac{1}{c^2}\int_{-\infty}^{\infty} \frac{\partial \Phi}{\partial b}\,ds = \frac{GM}{c^2}\int_{-\infty}^{\infty} \frac{b\,ds}{(b^2+s^2)^{3/2}} = \frac{2GM}{bc^2}. \tag{16}$$

Open: the spatial half of the deflection

The deflection (16) is exactly half the general-relativistic value $4GM/(bc^2)$. This is not an error, and not an approximation to be improved within the present paper: $2GM/(bc^2)$ is precisely the value Einstein obtained in 1911 from gravitational time dilation alone, before the full theory. The missing half is the contribution of the spatial sector of the acoustic metric (§8) — the analogue here of spatial curvature — and its derivation belongs to Paper VII-b. The time-delay refraction of this section delivers the time half in full and transparently; it does not reach the spatial half, and does not pretend to.

6.4 Free fall, and refraction as the potential gradient

Light is a wave, and the front that tilts is its wavefront. A massive defect is also extended (§6.2), and it too has a front that tilts — not its external wavefront but its internal oscillation.

The defect oscillates internally at the realised rate $\omega = \omega_0 A$ (§2.4); across its extent A varies, so the internal oscillation runs slower on the side toward the mass. After a coordinate time t the internal phase $\Theta = -\omega_0 A(\mathbf{x})\, t$ is not uniform across the defect but carries a gradient

$$\mathbf{k} = \nabla\Theta = -\,\omega_0\, t\, \nabla A. \tag{17}$$

A phase gradient is a momentum $\mathbf{p} = \hbar\mathbf{k}$, and it grows with time, so the defect feels a force

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = -\,\hbar\omega_0\, \nabla A = -\,mc^2\, \nabla A = -\,m\, \nabla\Phi, \tag{18}$$

using $\hbar\omega_0 = mc^2$ and $A = 1 + \Phi/c^2$. This is free fall — the acceleration $-\nabla\Phi$ of §5, recovered here from the wavefront construction. The tilt that bends light bends matter too: for light the external front, for matter the internal oscillation, each tilting in the one gradient of A. It is here that the extended defect of §6.2 earns its place — a point particle would have no width for the phase to tilt across, and no free fall to show for it.

Light and matter are now described twice over, and the two descriptions are one. Section 5 read free fall and deflection off the potential Φ; this section reads them off the refraction of a front in the graded availability A. They are the same statement: the ray curvature is $|\nabla_\perp\Phi|/c^2$, the free-fall acceleration is $|\nabla\Phi|$, and both are the gradient of the one potential. "Refraction in the time-delay field" and "motion down the gradient of Φ" are not two pictures of gravity but one — the wavefront construction is what the gradient $-\nabla\Phi$ does, mechanically, to anything that moves.

Numerical check. The refraction picture has been tested directly. A Gaussian wave packet is propagated past a softened potential well by a split-step solver and its deflection compared with the classical $\mathbf{F} = -\nabla\Phi$ trajectory through the same well. The measured deflection converges to the $-\nabla\Phi$ value as the packet is made point-like: the relative error falls from 6.5% to 1.3% as the ratio of impact parameter to packet width grows from 4 to 13. The residual is precisely the finite-width effect of §6.2 — an extended object samples a convex gradient and deflects a little more than a point at its centroid. Throughout the crossing the transverse phase tilt that develops across the packet tracks its transverse momentum, which is the mechanism of (18) itself: a phase gradient *is* a momentum. The run parameters and convergence tests are collected in Appendix A.

7 Universal Coupling and the Absence of a Graviton

7.1 Universal Coupling: Why Everything Falls

The interference mechanism is universal because every defect in the medium — regardless of internal structure, charge, or topology — radiates a wave, and any wave interferes with any other. The amplitude of that wave is set by the breather's mass-energy alone — the fractional volume excursion is $\delta V \propto m/\rho_0$ (Paper II) — so the interference force scales as $m_1 m_2/r^2$ for any pair of objects.

This is the mechanical origin of the equivalence principle: the same quantity (mass-energy) determines both the inertial response to acceleration and the coupling to gravity itself, because both are manifestations of the object's acoustic cross-section in the medium. Every other force is selective — electromagnetism couples to chirality, the strong force to topological charge, the weak force to core overlap — but gravity couples to the one property all defects share: they displace the medium.

7.2 Gravity Without a Graviton

The SSV framework does not require a graviton. The gravitational potential Φ is the time-averaged interference of §4; its $1/r$ long-range form is the spherical dilution of a radiated wave, equivalently the Green's function of the Laplacian. What conventional quantum field theory calls graviton exchange corresponds here to the interference of waves in the medium — a collective hydrodynamic process, not a particle interaction.

This is structurally consistent with the treatment of the other forces in Paper II: electromagnetism emerges from the transverse flow of chiral vortices without postulating a separate photon field, and gravity emerges from longitudinal density waves without postulating a separate graviton field. Both are modes of the single medium Ψ.

The ultraviolet problem dissolves. Conventional perturbative quantum gravity is non-renormalisable: when gravity is treated as a quantum field theory in the standard way, the loop integrals diverge at short distances and the infinities cannot be removed by redefining a finite number of parameters. The theory has no built-in short-distance cutoff and breaks down at the Planck scale, requiring replacement by an unknown theory of quantum gravity.

In SSV this problem does not arise — not because we solve it, but because the framework's domain of validity makes it impossible to state. SSV is a continuum effective theory valid down to the scale of the heaviest stable defect,

$$\ell_{\text{grain}} \equiv a_p = \frac{\hbar}{m_p c} \approx 2.10 \times 10^{-16}\,\text{m}. \tag{19}$$

Below ℓ_{grain}, the LogSE continuum description ceases to be self-consistent: no new stable structures exist, and the medium supports only transient denting events (Paper II). Momentum modes with $k \gg 1/\ell_{\text{grain}}$ are outside the framework's domain of validity — not regulated away by hand, but simply absent from the theory, in the same sense that Navier–Stokes has no sub-molecular eddies. The divergent integrals of perturbative quantum gravity never appear because the modes that would generate them are outside the scope of the description.

This is worth pausing on. The ultraviolet catastrophe of quantum gravity is usually treated as one of the deepest unsolved problems in physics, motivating decades of work on string theory, loop quantum gravity, and related programmes. In SSV it is not solved — it simply does not arise. The cutoff $\ell_{\rm grain}$ is not a new postulate introduced to tame divergences; it is the healing length of the heaviest stable defect, fixed by the proton mass alone. The same physical reasoning that gives Navier–Stokes its molecular cutoff gives SSV its cutoff at a_p: at scales finer than the smallest stable structure, the continuum description is silent.

Note the distinction from the electron healing length $\xi = \hbar/(m_e c) \approx 3.86 \times 10^{-13}$ m, the natural scale for electron-defect physics but not the medium's ultraviolet boundary: the true domain-of-validity boundary is $\ell_{\rm grain} \approx \xi/1836$, set by the proton.

The absence of a graviton and the absence of UV divergences are the same fact stated two ways: SSV has no gravitational field to quantise, and therefore no quantisation problem to solve.

8 From Update-Capacity Gradients to Emergent Geometry

The time-delay gradient of the preceding sections is the weak-field face of a deeper statement: that geometry itself is a macroscopic limit of the medium. This section sketches the bridge; its full development — the effective metric to all orders, the post-Newtonian coefficients, and why the standard quantum-gravity problem is mis-posed in SSV — is the subject of Paper VII-b.

The curved-spacetime formulation of gravity emerges from the Madelung equations in the hydrodynamic limit [2, 3]. The acoustic metric

$$g_{\mu\nu}^{\rm eff} = \frac{\rho_0}{m_0 c}\begin{pmatrix} -(c^2 - v^2) & -v_j \\ -v_i & \delta_{ij} \end{pmatrix} \tag{20}$$

reduces to Minkowski when the background flow vanishes.

Its time component is fixed by the results of this paper. The availability A of §2.4 is the rate of proper time, $d\tau/dt = A$; since a clock at rest measures $d\tau = \sqrt{-g_{00}}\,dt$, the time–time component of the weak-field metric is

$$-g_{00} = A^2 = \left(1 + \frac{\Phi}{c^2}\right)^2 \approx 1 + \frac{2\Phi}{c^2}, \tag{21}$$

the standard weak-field g_{00} — its factor of two present in full, read straight off the update budget rather than postulated. The time sector of the metric is thus delivered complete. What this paper does *not* reach is the spatial sector g_{ij}: it is the geometrical face of the emergent Lorentz structure of §2.3, and its derivation — which supplies the spatial half of light deflection and so completes (16) to the full $4GM/(bc^2)$ — is the work of Paper VII-b.

A single potential Φ governs both sectors: it sets the clock rate $1 + \Phi/c^2$ through g_{00} and the remaining wave deflection through g_{ij}. That one potential does both — the force $-\nabla\Phi$ and the clock rate — is precisely the weak-field structure of general relativity. Gravitational lensing follows: light is a wave in the same Ψ, deflected by the same gradient that bends the path of a test mass. No separate gravitational field is postulated at any point.

9 Gravity in a Medium That Remembers

The treatment above is the mean-field story. Two specific places in the derivation depend on operations that the irreversibility framework of Paper III will revisit at finer resolution; they are flagged here so that the present scope is honest.

Time-averaging the interference. Newton's constant was derived in Paper II from the time-averaged interference cross-term $\langle F \rangle = -\rho_0 \langle \dot{V}_1 \dot{V}_2 \rangle / (4\pi r^2)$. The averaging brackets perform a classical operation: integrate over many cycles of the proton breather oscillation at $\omega_p = m_p c^2/\hbar$ to extract the static component. In a featureless medium this is unproblematic. In a medium that records, what gets averaged is itself history-dependent: each cycle deposits a small amount of acoustic energy into the medium, and the breather of cycle $n+1$ propagates through a slightly different background than the breather of cycle n. The leading-order static result survives this refinement — the recoil of one cycle on the next is suppressed by the proton's sub-grain coupling factor $(m_e/m_p)^3$, the same factor that makes G small in the first place — but the corrections are non-zero and carry information about the source's pulsation history. At laboratory scales the suppression is severe: $(m_e/m_p)^3 \approx 1.6 \times 10^{-10}$, so the history-dependent corrections stand ten orders of magnitude below the leading gravitational effect — far beneath the precision of any gravity measurement now conceivable, and safely inside the mean-field treatment of this paper. These corrections are the gravitational-wave memory effect (Christodoulou) read in this framework: not as a peculiar property of asymptotic spacetimes, but as the direct consequence of a medium that literally remembers.

The interference pattern as a long-time average. The static potential Φ is the long-time-averaged envelope of the interfering pulsation waves. In the framework's deeper picture, $\rho(\vec{x}, t)$ is not a smooth deterministic function but a coarse-grained description of an evolving record; the smoothness emerges because reconnection events and their acoustic wakes are dense enough to wash out at scales relevant to the equivalence-principle tests we currently perform. At sufficiently high precision and sufficiently short timescales, this smoothing must fail: the gravitational potential at a point will fluctuate around its mean value at a rate set by the local rate of reconnection events. No calculation of this fluctuation spectrum is given here. It is identified as the natural Paper III follow-up to the present mean-field gravity treatment.

What this does not change. The Newton's-constant consistency check of Paper II is unaffected: it relates time-averaged quantities, all defined at the same level of coarse graining. The graviton-free argument is unaffected: there is still no gravitational field to quantise. The argument that ℓ_{grain} supplies the natural ultraviolet cutoff is unaffected: it remains the medium's minimum length. What the forward connection makes explicit is only that the time-averaging step inherits the medium's accumulated history at a level the present treatment does not resolve — and that this is not an obstacle to the mean-field claims, only a scoping of where Paper III takes over.

10 Discussion and Open Problems

The mechanism set out here is complete at leading order and introduces no parameter beyond those fixed in Paper I. Gravity is the interference of the pressure waves that breathers radiate into the one medium; the interference slows local time; its gradient bends motion. From the single potential Φ the four classical gravitational phenomena follow without further assumption, and the same logarithmic nonlinearity that makes the medium a medium also makes gravity attractive and inverse-square.

What is established: the interference cross-term (9) and its three structural properties; the four corollaries of §5 at $O(\Phi/c^2)$; the absence of a graviton and of an ultraviolet divergence. What is deferred: the first-principles value of the gravitational coupling — the sub-grain factor that fixes G — remains the open computation of Paper II; the extension beyond leading order, and with it the post-Newtonian coefficients and the full emergent metric, is Paper VII-b. Sections 2 and 6 — the formal account of update capacity and the refraction geometry — are now complete, so the paper delivers the time sector of gravity in full at $O(\Phi/c^2)$. Its one half-strength result, light deflection, is half-strength by design: light draws equally on the time and spatial sectors, and the spatial sector is Paper VII-b, not a gap in the present argument. Both ends of the mechanism have now been checked against direct simulation: a split-step propagation of a wave packet past a potential well recovers the free-fall acceleration $-\nabla\Phi$ in the point-like limit (§6.4), and a simulation of two pulsating sources in the logarithmic medium recovers the bilinear, sign-fixed interference cross-term of §4, attractive for phase-correlated waves. At leading order the mechanical account set out here is now a tested one.

A Numerical Methods and Convergence

The two numerical checks of §4 and §6 share a method: a two-dimensional split-step Fourier integrator with Strang operator splitting — a half-step of the potential, a full kinetic step applied exactly in Fourier space, then a second half-step of the potential — in units $\hbar = m = 1$. This appendix records the parameters of each run and the grid- and timestep-convergence behind the numbers quoted in the text. The solvers are `wave_deflection.py` and `breather_interaction.py`; each carries a `conv` mode that reproduces the convergence tables below.

A.1 The wave-deflection check (§6)

The solver propagates a Gaussian wave packet across the softened well $V = -\alpha/\sqrt{r^2+\epsilon^2}$ under $i\,\partial_t\psi = -\frac{1}{2}\nabla^2\psi + V\psi$, and reads the deflection off the momentum expectation $\langle \mathbf{p}\rangle$ evaluated in Fourier space. The classical $\mathbf{F} = -\nabla\Phi$ trajectory through the same well, integrated by velocity-Verlet, is the reference. Table 1 lists the parameters.

Quantity	Symbol	Value
Grid points per axis	N	512
Box size	L	200
Grid spacing	Δx	0.39
Timestep	Δt	0.025
Packet width	σ	4
Packet speed (carrier $k = v$)	v	2.5
Well depth	α	2
Well softening	ϵ	4
Initial centroid offset	x_0	-65
Impact parameters	b	16, 24, 36, 52
Classical reference	—	velocity-Verlet, 4×10^4 steps

Table 1: Wave-deflection solver parameters. The packet is carried from $x = x_0$ to $x = -x_0$, a propagation time $T = 2|x_0|/v$.

The point-limit result of §6 — the relative error between the measured deflection and the classical value falling from 6.5% to 1.3% as b/σ grows from 4 to 13 — is a *physical* convergence: a finite-width packet samples a convex gradient and deflects slightly more than a point at its centroid. That this is physics, not discretisation error, is shown by Table 2: at the fixed case $b/\sigma = 6$ the measured deflection does not move in any printed digit under a threefold grid refinement or a fourfold timestep reduction. The residual 4.81% at this case is entirely the finite-width effect, and the `run_case` defaults ($N = 512$, $\Delta t = 0.025$) sit well inside the converged regime.

Refinement		Measured deflection	Rel. error
Grid ($\Delta t = 0.025$)	$N = 256$	-0.02557	4.81%
	$N = 384$	-0.02557	4.81%
	$N = 512$	-0.02557	4.81%
	$N = 768$	-0.02557	4.81%
Timestep ($N = 512$)	$\Delta t = 0.0500$	-0.02557	4.81%
	$\Delta t = 0.0250$	-0.02557	4.81%
	$\Delta t = 0.0125$	-0.02557	4.81%

Table 2: Wave-deflection convergence at the fixed case $b = 24$, $\sigma = 4$. The classical $-\nabla\Phi$ reference is -0.02439; the measured deflection is invariant to all printed digits.

A.2 The breather-interaction check (§4)

The solver integrates the logarithmic-Schrödinger equation $i\,\partial_t\psi = -\frac{1}{2}\nabla^2\psi + b\ln|\psi|^2\,\psi + V(\mathbf{x},t)\,\psi$ on the uniform background $\psi = 1$. Two localised drives, $V(\mathbf{x},t) = V_0\,\mathrm{ramp}(t)\sin(\omega t)\,[W_A + s\,W_B]$ with $s = \pm 1$, pulse the medium; the time-averaged force on source A, $\langle F_{A,x}\rangle = \langle \int \rho\,\partial_x V_A\,d\mathbf{x}\rangle$, is accumulated over an integer number of drive periods. A relaxation absorber, $\psi \to 1 + (\psi - 1)\,e^{-\Gamma\Delta t}$ in a layer at the box edges, removes outgoing waves so the measurement is of radiation, not of box modes. The nonlinearity $b = 1$ fixes the sound speed $c_s = \sqrt{b} = 1$ and hence the wavelength $\lambda = 2\pi c_s/\omega$. Table 3 lists the parameters.

Quantity	Symbol	Value
Grid points per axis	N	160
Box size	L	100
Grid spacing	Δx	0.625
Timestep	Δt	0.025
Nonlinearity	b	1
Sound speed	$c_s = \sqrt{b}$	1
Drive amplitude	V_0	0.25
Source width	w	1.6
Drive frequency, headline / sweep	ω	0.7 ($\lambda \approx 9$)
Drive frequency, near-zone	ω	0.15 ($\lambda \approx 42$)
Absorber width	—	18
Absorber strength	$\Gamma_{\max}$	1.5
Transient before averaging	—	70
Averaging window	—	8 drive periods

Table 3: Breather-interaction solver parameters. The near-zone sweep lowers ω to push λ well beyond the separation; its ramp, transient and averaging window are scaled to the longer drive period.

The headline run reports an isolated source feeling a residual force of 3.6×10^{-5} — three orders below the in-phase forces of order 3×10^{-2} — an in-phase to anti-phase ratio of -1 to within 1% at every separation, near-zone attraction, and a radiation-zone force that oscillates in sign with period λ under an envelope $\sim 1/\sqrt{d}$, the two-dimensional cylindrical-wave law.

Table 4 gives the convergence: at the fixed in-phase case $d = 20$ the measured force agrees to four significant figures across grids from $N = 128$ to $N = 256$ and under a halving of the timestep, so the $N = 160$, $\Delta t = 0.025$ defaults of the sweeps are firmly in the converged regime.

Refinement		$\langle F_{A,x} \rangle$
Grid ($\Delta t = 0.025$)	$N = 128$	$+3.28685 \times 10^{-2}$
	$N = 160$	$+3.28675 \times 10^{-2}$
	$N = 192$	$+3.28681 \times 10^{-2}$
	$N = 256$	$+3.28683 \times 10^{-2}$
Timestep ($N = 160$)	$\Delta t = 0.0250$	$+3.28675 \times 10^{-2}$
	$\Delta t = 0.0125$	$+3.28728 \times 10^{-2}$

Table 4: Breather-interaction convergence at the fixed in-phase case $d = 20$, $\omega = 0.7$. The force is stable in its fourth significant figure under both refinements.

References

[1] V. F. K. Bjerknes, *Fields of Force*, Columbia University Press, 1906.

[2] G. E. Volovik, *The Universe in a Helium Droplet*, Oxford University Press, 2003.

[3] C. Barceló, S. Liberati, and M. Visser, "Analogue gravity," *Living Rev. Relativity* **14**, 3 (2011).